

Functional rescue of F508del-CFTR through revertant mutations introduced by CRISPR base editing

Irene Carrozzo,^{1,7} Giulia Maule,^{1,7} Carmelo Gentile,¹ Alessandro Umbach,¹ Matteo Ciciani,¹ Daniela Guidone,² Martina De Santis,² Gianluca Petris,^{1,5,6} Luis Juan Vicente Galiotta,^{2,3} Daniele Arosio,^{4,8} and Anna Cereseto^{1,8,9}

¹Department CIBIO, University of Trento, Via delle Regole 101, 38123 Trento, Italy; ²Telethon Institute of Genetics and Medicine, 80078 Pozzuoli, Italy; ³Department of Translational Medical Sciences, University of Napoli “Federico II”, 80138 Napoli, Italy; ⁴Institute of Biophysics, CNR, Via alla Cascata 56/C, 38123 Trento, Italy

Cystic fibrosis (CF) is a life-shortening autosomal recessive disease caused by mutations in the *CFTR* gene, resulting in functional impairment of the encoded ion channel. F508del mutation, a trinucleotide deletion, is the most frequent cause of CF, affecting approximately 80% of persons with CF (pwCFs). Even though current pharmacological treatments alleviate the F508del-CF disease symptoms, there is no definitive cure. Here, we leveraged revertant mutations (RMs) in *cis* with F508del to rescue CFTR protein folding and restore its function. We developed CRISPR base editing strategies to efficiently and precisely introduce the desired mutations in the F508del locus. Both editing and CFTR function recovery were verified in CF cellular models, including primary epithelial cells derived from pwCFs. The efficacy of the CFTR recovery strategy was validated in cultures of pseudostratified epithelia from pwCF cells showing full recovery of ion transport. Additionally, we observed an additive effect by combining our strategy with small molecules that enhance F508del activity, thus paving the way to combinatorial therapies.

INTRODUCTION

Cystic fibrosis (CF) is a life-shortening autosomal recessive disease that affects at least 100,000 people worldwide.^{1,2} It is caused by mutations in the CF transmembrane conductance regulator (*CFTR*) gene, which encodes a transmembrane channel localized at the apical surface of the epithelial cells.³ CFTR is fundamental to maintaining correct homeostasis across epithelial membranes by ensuring mucus hydration and clearance, in particular, in the airways. Indeed, CFTR impairment affects multiple organs, yet the morbidity and mortality of persons with CF (pwCFs) are caused mainly by progressive obstructive lung disease.^{1,2}

According to the CFTR2 database,⁴ more than 700 mutations in the *CFTR* gene are causative of CF. The most common mutation is the F508del (present in 80% of pwCFs), a deletion of 3 nt that determines the loss of a phenylalanine residue.^{2,4} This genetic alteration results in a misfolded CFTR protein that is targeted for degradation by the ubiquitin-proteasome system.⁵ A small fraction of mutated CFTR can evade early degradation and reach the plasma membrane (PM),

but it remains insufficiently active and unstable. Efficient CF therapies were developed consisting of small molecules that modulate CFTR folding (correctors) or gating activity (potentiators).² The current recommended treatment for pwCFs carrying the F508del mutation (in one or both alleles) is a combination of three modulators, elexacaf-tor-tezacaftor-ivacaftor (ETI), consisting of two correctors and a potentiator.² Although ETI significantly improves the health of pwCFs, it is not a definitive cure and requires lifelong administration. As often occurs with pharmacological regimens, these treatments can cause side effects and occasionally may not produce the desired therapeutic efficacy due to various factors, including genetic variability.^{6,7}

Genome editing is a new powerful instrument for gene therapy to treat genetic diseases.⁸ In CF, various genome editing strategies have been developed to repair CFTR mutations, including the F508del defect.⁹ Given the nature of this mutation being a small deletion, several groups used the homology-directed repair (HDR) strategy allowing the replacement of sequences favored by programmable nucleases, including zinc-finger nucleases or Cas9.^{10–15} However, this method is limited to specific target cells as it depends on specific phases of the cell cycle (S-G2/M) and is associated with risks of genotoxicity induced by the DNA double-strand breaks (DSBs).^{16–18} Recent technological advancements in genome editing include the development of prime editing,¹⁹ a DSB-free approach that was efficiently used to correct F508del reaching up

Received 6 September 2024; accepted 8 January 2025;
<https://doi.org/10.1016/j.ymthe.2025.01.011>.

⁵Present address: Fondazione Italiana Fegato ONLUS - Italian Liver Foundation NPO, 34149 Trieste, Italy

⁶Present address: Department of Medicine, Università degli Studi di Udine, 33100 Udine, Italy

⁷These authors contributed equally

⁸These authors contributed equally

⁹X (formerly Twitter): @CIBIOcrispr

Correspondence: Daniele Arosio, Institute of Biophysics, CNR, Via alla Cascata 56/C, 38123 Trento, Italy.

E-mail: daniele.arosio@cnr.it

Correspondence: Anna Cereseto, Department CIBIO, University of Trento, Via delle Regole 101, 38123 Trento, Italy.

E-mail: anna.cereseto@unitn.it

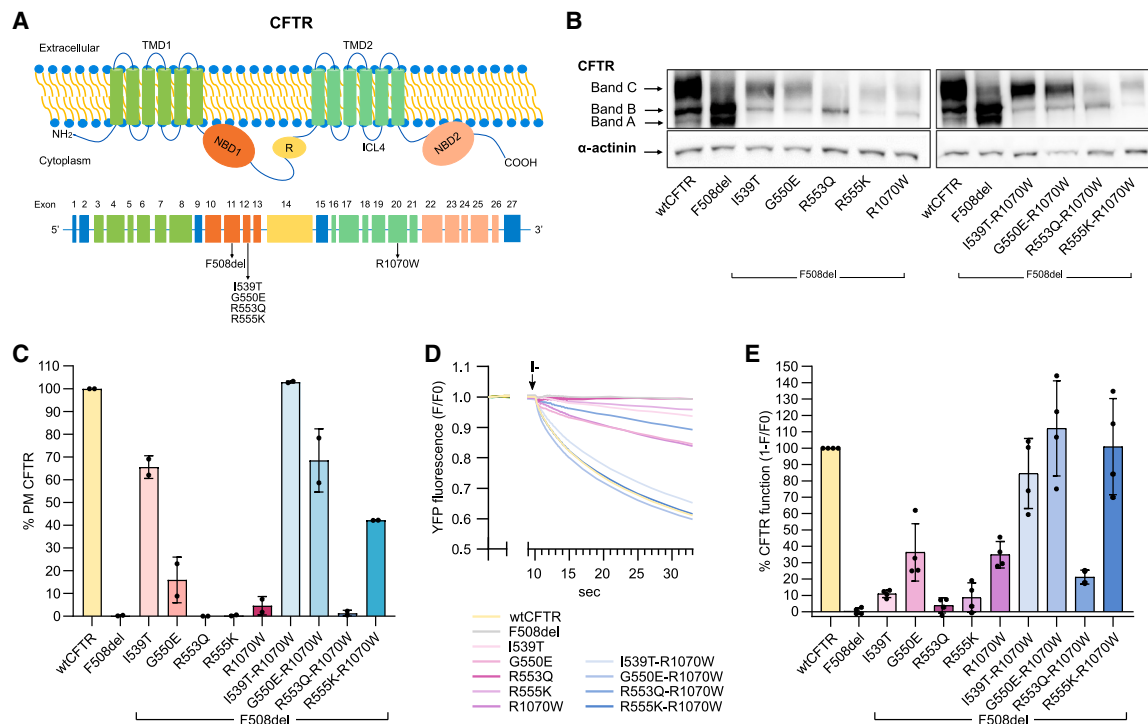


Figure 1. RMs rescue F508del-CFTR maturation, localization, and function

(A) Schematic representation of CFTR protein structure, highlighting its five domains: two transmembrane domains (TMD1 and TMD2), each formed by six transmembrane segments, two cytosolic nucleotide binding domains (NBD1 and NBD2), and the regulatory domain (R). (Bottom) Exons and the locations of the F508del and RMs within the CFTR locus. Corresponding protein domains are color coded. (B) Representative western blot analysis of the F508del modified by the indicated RMs and transduced in HEK293T cells with lentiviral vectors. The blot was developed with anti-CFTR antibodies (top) or α -actinin (bottom). Bands A, B, and C are indicated by the arrows. (C) Histograms representing the percentage of CFTR (3HA-tagged) rescued at the PM detected by flow cytometry. Samples are normalized to wtCFTR, which is considered 100% at the PM. Data are means $\pm$ SDs from $n = 2$ independent experiments. Gating strategy is shown in Figure S1A. (D and E) HS-YFP quenching assay performed in HEK293T stably expressing HS-YFP and transfected with the indicated CFTR variants. (D) Representative traces of HS-YFP assay recorded with a fluorescence microscope (see methods). Cells were stimulated with 10 μ M forskolin before the addition of iodide (I^-), indicated by the arrow. (E) Quantification of CFTR activity as a measure of HS-YFP quenching from $n = 4$ independent experiments. CFTR function is calculated as $1 - F/F_0$ using values recorded 7 s after I^- addition and normalized on wtCFTR sample. Data are means $\pm$ SDs.

to 50% function compared to wild type.^{20,21} Nonetheless, prime editing generates a high percentage of unwanted small insertions/deletions (indels) with unpredicted outcomes,^{20,21} and its composition with multiple elements increases the molecular size of the delivery cargo, thus challenging the *in vivo* treatments.^{22,23}

Previous studies showed that revertant mutations (RMs) in *cis* with F508del can stimulate the proper folding and function of CFTR protein.²⁴ The first RM was identified in 1991 with a pwCF reporting a milder form of CF.²⁵ The genetic analysis revealed a compound F508del-R553Q and R553X, the latter corresponding to a nonsense mutation. Therefore, the milder phenotype was very likely generated by the second-site mutation in the F508del-CFTR locus which at least partially abrogated the CF phenotype. This observation prompted the search for additional RMs, leading to the identification of I539T, G550E, R555K, and R1070W.^{26–32} Most of these RMs are located within the nucleotide binding domain (NBD1) of CFTR, except for the R1070W, which resides in the fourth intracellular loop (ICL4)

(Figure 1A). RMs in the NBD1 reduce the thermodynamic and kinetic instability during F508del-CFTR biogenesis,^{29–31} while R1070W restores the ICL4-NBD1 interface.^{30,32}

Here, we exploit CRISPR base editors to introduce RMs into the F508del-CFTR locus as a strategy to restore CFTR activity. The efficiency, precision, and competitive molecular size among the genome editing tools offer the opportunity for a definitive cure to F508del pwCFs.

RESULTS

Restoration of F508del defect mediated by RMs in CF cell lines

We evaluated the impact of RMs I539T, G550E, R553Q, R555K, and R1070W on F508del-CFTR by first analyzing the protein maturation. Different CFTR maturation products are generated during the protein channel biogenesis, which can be detected through western blot analysis: a non-glycosylated form, band A (135 kDa); an immature partly glycosylated form, band B (150 kDa); and a fully glycosylated mature form, band C (170 kDa).³³ The F508del-CFTR carrying

the RMs, single or in combination, were expressed in HEK293T cells using lentiviral vectors. The I539T-R1070W and G550E-R1070W induced the formation of mature CFTR proteins at levels comparable to wild-type CFTR, while partial maturation rescue was observed with the single I539T, G550E, R555K, R1070W, and combined R555K-R1070W RMs (Figure 1B).

To further evaluate the rescue efficacy of the RMs on F508del-CFTR, we analyzed CFTR cellular localization, which is physiologically at the PM if not impaired by mutations that alter protein folding.³⁴ We performed flow cytometry analyses using a triple hemagglutinin (3HA)-tag located at the fourth external loop of CFTR. The amount of CFTR restored at the PM (Figures 1C and S1A) mirrored the levels of maturation observed by western blot, showing a strong localization rescue by the combinations I539T-R1070W and G550E-R1070W and also by the I539T, while the remaining mutations showed variably lower levels of rescue.

Finally, to assess the CFTR function, we performed the halide-sensitive yellow fluorescent protein (HS-YFP) assay^{35,36} in HEK293T cells. This assay exploits the quenching property of HS-YFP by iodide anions. In cells where CFTR is functionally active, iodide ions enter the cells through the CFTR channels, leading to the quenching of HS-YFP, which results in fluorescence decrease, while in CFTR⁻ cells the fluorescence remains unaltered. This assay was performed in HEK293T stably expressing HS-YFP and transfected with various F508del-CFTR variants. Consistent with the maturation and cellular localization data, the I539T, G550E, and R555K combined with R1070W rescued CFTR activity to near wild-type levels. Partial or no rescue was observed with the single revertants and the combination R553Q-R1070W (Figures 1D and 1E).

CRISPR-base editing strategies to introduce RMs in F508del-CFTR

CRISPR-base editor strategies were developed to introduce RMs I539T, G550E, R553Q, R555K, and R1070W in the F508del-CFTR locus (Figure S1B). We used HEK293T cells carrying the F508del-CFTR cDNA to monitor the incorporation of specific RMs and to assess any nucleotide modifications occurring beyond the primary target, known as bystander edits.³⁷

Depending on the RM to be inserted, we used either adenine base editors (ABEs)³⁸ or cytosine base editors (CBEs)³⁹ for A>G or C>T substitutions, respectively, utilizing different versions of nucleotide deaminases.^{40–42} By leveraging the highest compatibility with protospacer adjacent motif (PAM) recognition sequences surrounding the target sites, we tested base editors made with three SpCas9 variants, SpCas9-NG,⁴³ SpG, and SpRY.⁴⁴ For each base editor, several single-guide RNAs (sgRNAs) were tested, indicated with numbers corresponding to the position of the target nucleotide relative to the PAM sequence⁴⁵ (Figures S2A–S2E).

To insert the I539T mutation, we tested several ABE versions (ABE8e, ABE8.20m, and ABEmax).^{40–42} We obtained variable levels of A>G conversions at the target nucleotide, with up to 45% of editing with

SpG-ABEmax, SpG-ABE8.20m, and SpG-ABE8e (Figure 2A). Most of the ABEs tested showed substantial bystander activity in positions A4 and A9, except for NG-ABEmax and SpG-ABEmax, which showed minimal conversion at A4 and less than 10% conversion at A9, resulting in a silent N537N mutation.

To insert G550E, R553Q, R555K, and R1070W mutations, we used two CBE versions, BE4max and AncBEmax,⁴¹ combined with different sgRNAs and SpCas9 variants (Figures 2B–2E).

For the G550E mutation, we obtained up to 45% editing at the target cytosine (C5), associated with relevant bystander edits throughout the base editing window (Figure 2B). Two specific bystander edits, C2 and C8, are predicted to have a major functional impact since they produce CF-causing mutations (G551D and S549N)⁴⁶ (Figure 2B).

The R553Q was introduced, with up to 50% efficiency at C6 using various combinations of CBEs and sgRNAs (Figure 2C) showing main bystander edits at C15 (Figure 2C). Interestingly, the C15 bystander introduces another RM, the G550E, which is beneficial in the rescue of the F508del-CFTR folding. The most efficient editing strategies, as a result of both specific and synergistic bystander editing, were obtained with SpRY-BE4max and SpG-BE4max with sgRNA+6, generating the simultaneous introduction of G550E-R553Q RMs.

The R555K was generated with SpRY-BE4max in combination with two different sgRNAs. Target base conversion reached over 50% efficiency with sgRNA+5 (Figure 2D), with the advantage of introducing a second RM, the R553Q through a bystander edit in C11, thus resulting in double R553Q-R555K RMs. The R1070W mutation was introduced using SpRY-BE4max and SpG-AncBE4max, resulting in up to 30% editing at C5. Additionally, a bystander edit at C2 generated a silent mutation, the F1068F (Figure 2E).

Next, we explored the possibility of using a single base editor to introduce combinations of RMs. The possible compatible combinations were R553Q-R555K, R553Q-R1070W, and R555K-R1070W, while I539T and G550E could not be inserted together with others. The R553Q-R555K, R553Q-R1070W, and R555K-R1070W were inserted using SpRY-BE4max, while the R553Q-R1070W was inserted with SpG-AncBE4max. The RM combinations were overall efficiently edited (Figure 2F). We observed a more evident reduction of R555K editing when combined with the R553Q, probably due to sgRNAs interference (Figure 2F). Bystander edits were similar to those of single-guide strategies (Figure S3).

Rescue of F508del-CFTR with RMs introduced via base editing

We evaluated the rescue of the F508del-CFTR modified by using the most efficient base editing strategy identified in Figure 2. We analyzed PM localization, which showed partial rescue (5%–31% compared to wild type) with all RMs (Figure 3A). Interestingly, the individual R553Q and R555K showed higher PM recovery than the corresponding modifications in F508del-CFTR cDNA (compare with Figure 1C). The higher PM recovery was probably due to bystander editing

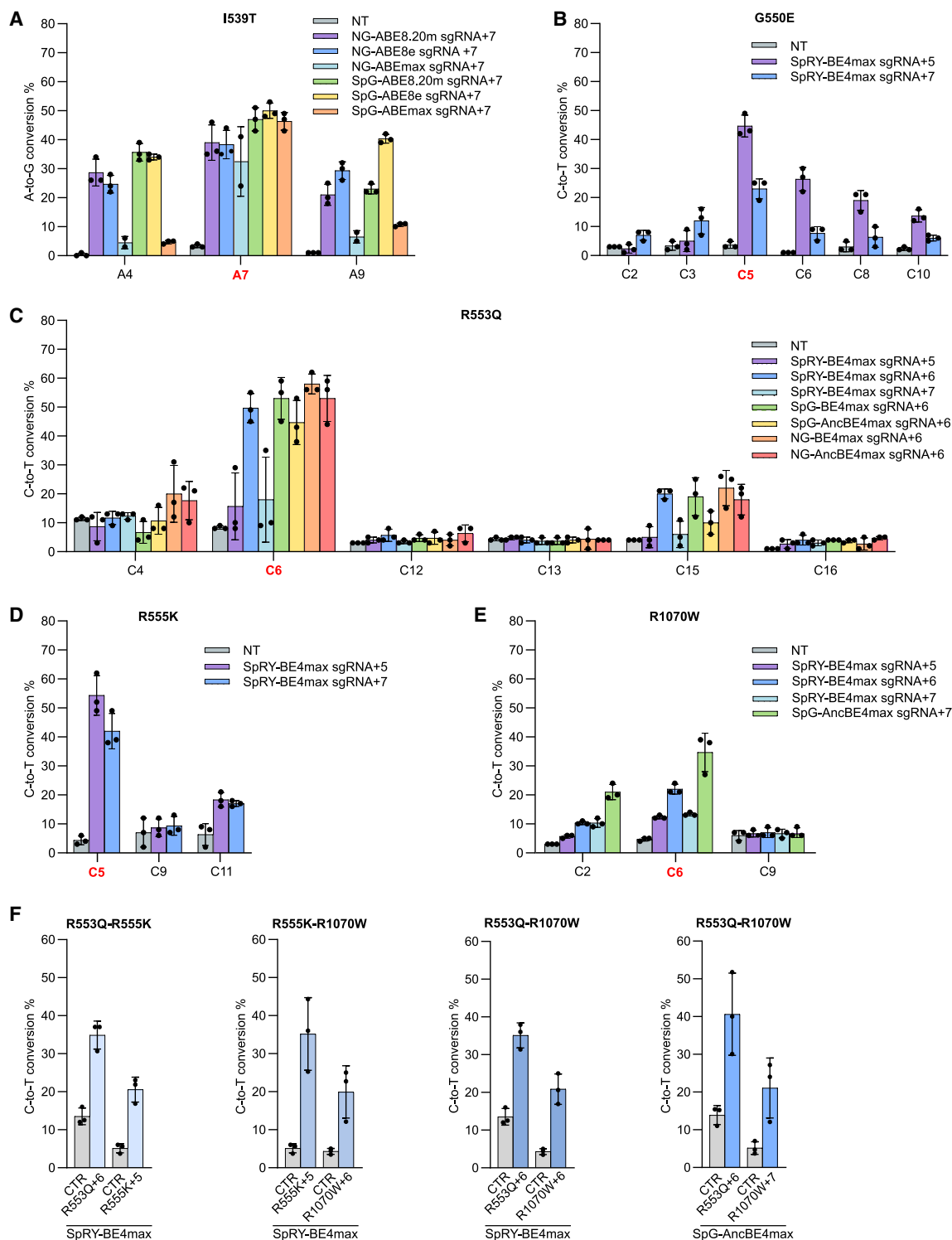
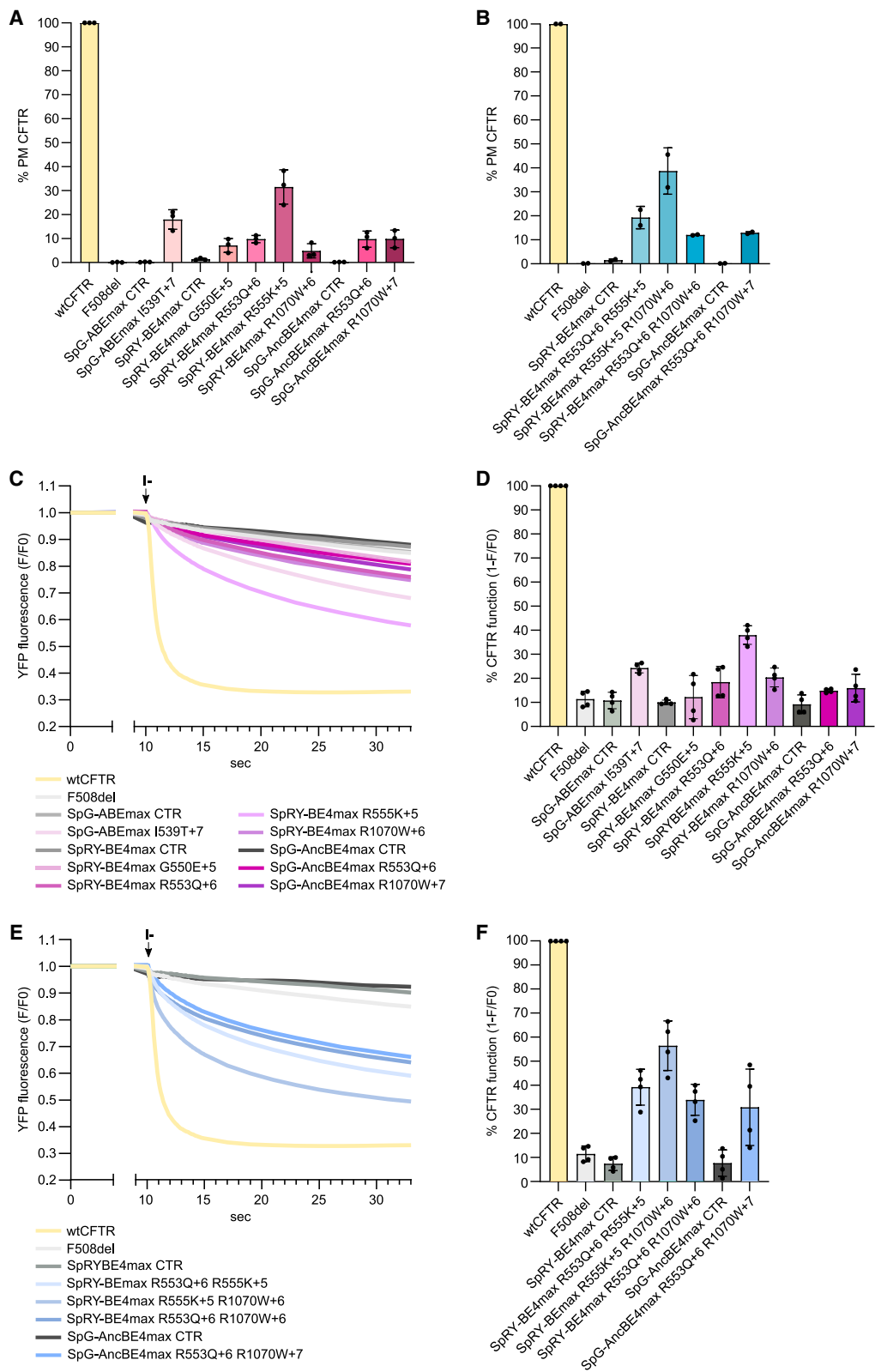


Figure 2. Insertion of RMs in the F508del-CFTR locus through CRISPR base editors

Editing efficiency in F508del-CFTR HEK293T cells transfected with the indicated base editors and sgRNAs used (A–E) individually or (F) in combination to generate RMs indicated at each graph. Editing was quantified by EditR tool⁴³ after PCR amplification and Sanger sequencing of the target locus and compared with nontreated (NT) cells. The target base for each RM is highlighted in red; bystander edits are shown in black. Data are means ± SDs from $n = 3$ independent experiments.



(legend on next page)

introduced by the additional RMs, as reported in [Figures 2C and 2D](#). We also assessed PM localization with combinations of RMs which showed higher CFTR rescue, with the maximum generated by R555K-R1070W reaching 40% of the wild-type level ([Figure 3B](#)). The R553Q-R555K RMs showed more decreased PM localization than the R555K introduced individually ([Figures 3A and 3B](#)), probably due to interference between the sgRNAs (see [Figures 2D and 2F](#)).

CFTR function was then evaluated through HS-YFP quenching assay. Among single RMs, R555K showed the highest activity (up to 40%) ([Figures 3C and 3D](#)), while with the double RMs, R555K-R1070W showed up to 56% of wild-type activity ([Figures 3E and 3F](#)), paralleling the results observed with PM analysis.

Combinatorial rescue of F508del-CFTR with RMs and CFTR modulators

We investigated the potential synergism between RMs and pharmacological treatments, including the corrector lumacaftor, which enhances CFTR folding, and the potentiator ivacaftor, which improves CFTR gating. To this end, we performed the HS-YFP quenching assay in HEK293T cells stably expressing the variants of F508del-CFTR cDNA carrying the RMs.

We observed that individual RMs enhanced the activity of F508del-CFTR when treated with either modulators alone or in combination ([Figures 4 and S4](#)), with the exception of R553Q, where no synergism was observed. For the majority of combined RMs, I539T-R1070W, G550E-R1070W, and R555K-R1070W, which already increased F508del-CFTR activity close to wild-type levels, we detected only a modest improvement in CFTR function when modulators were added. Conversely, RM combinations with lower baseline activity, R553Q-R555K and R553Q-R1070W, synergized with both modulators, reaching wild-type levels of activity.

Efficiency and precision of F508del-CFTR base editing in patient-derived primary bronchial epithelial cells

The base editing strategy was tested in CF models, consisting of cultures of human primary bronchial epithelial (HBE) cells derived from a patient homozygous for the F508del mutation. Base editors were delivered by electroporation of RNA for both mRNA and sgRNA modified for optimal efficacy (see [methods](#)) and by using specific ratios ([Figures S5A and S5B](#)). The efficacy of F508del modification with single or double RMs was as efficient as in the HEK293T models ([Figures 5A–5E](#)). RNA delivery of base editors reduces bystander activity as previously reported,⁴⁵ likely due to transient expression. This reduction was not observed for the I539T mutation, which showed off-target

(OT) editing at A9, nevertheless leading to the generation of a silent mutation ([Figure 5A](#)), as observed in HEK293T cells ([Figure 2A](#)).

To assess the editing specificity, as reported before,^{45,48–50} we first performed a whole OT genome analysis (GUIDE sequencing [GUIDE-seq])⁵¹ with SpG or SpRY nucleases, followed by deep sequencing analysis of the nominated OTs in cells treated with base editors. A limited number of OT sites were revealed by SpG targeting the I539T locus ($n = 5$) or SpRY targeting R555K ($n = 1$) and R1070W ($n = 9$) loci. In contrast, a higher number of OTs ($n = 20$) were retrieved with the sgRNA R553Q+6 ([Figure S6](#)). The deep sequencing analyses to verify base editing on the retrieved OTs were performed in HBE cells by examining OTs with at least ≥ 30 sequencing reads in GUIDE-seq, thus largely over background levels. The percentage of base conversion reported in [Figure 5F](#) is the sum of all adenines or cytosines modified into the sgRNA spacer sequence. We measured 10% cumulative A>G conversion at OT1 with SpG-ABEmax I539T+7, corresponding to intron 1 of the *KCNT2* gene. SpRY-BE4max R553Q+6 showed three main OT sites, OT1, OT2, and OT4, among which the strongest was OT1, with a C>T conversion higher than the on-target site, corresponding to intron 3 of the *GALNT6* gene. Finally, SpRY-BE4max R1070W+6 exhibited almost 8% C>T conversion in OT2 corresponding to intron 1 of the *FLRT2* gene. No indels were detected beyond the reported A>G or C>T conversions ([Figure S7](#)).

Functional rescue of CFTR anion transport

Based on the editing efficiency and precision combined with the capacity of the RMs to rescue CFTR activity, we selected the R555K-R1070W RMs for further characterization to restore the F508del function. For this purpose, we analyzed terminally differentiated pseudostratified epithelia generated from HBE cells under air-liquid interface (ALI) conditions. To evaluate CFTR-dependent transepithelial ion transport, we measured short-circuit current (I_{sc}) in a Ussing chamber-like system.^{52,53}

In the first set of experiments, we blocked the sodium current mediated by ENaC with amiloride and then stimulated CFTR activity with the membrane-permeable cyclic AMP (cAMP) analog 8-(4-chlorophenylthio) adenosine 3',5'-cyclic monophosphate (CPT-cAMP). Subsequently, we added the selective CFTR blocker CFTR_{inh}-172 (I-172) to abolish CFTR current. CFTR-dependent chloride secretion, as indicated by the amplitude of the current elicited by CPT-cAMP and the extent of current drop resulting from I-172 addition, was larger in cells edited with the R1070W and R555K-R1070W compared to HBE cultures not treated (NT) or treated with a scramble sgRNA (CTR) ([Figures 6A and S8](#)). In particular, the drop caused by

Figure 3. Rescue of F508del-CFTR localization and function induced by insertion of RMs

(A and B) Percentage of CFTR at the PM measured by flow cytometry in wtCFTR and F508del-CFTR HEK293T cells transfected with the indicated base editors and sgRNAs producing (A) single or (B) double RMs or with a scramble sgRNA (CTR). Data are means $\pm$ SDs from $n = 3$ independent experiments. Samples are normalized to wtCFTR, which was set at 100%. (C–F) HS-YFP quenching assay in HEK293T endogenously expressing HS-YFP and F508del-CFTR cDNA and transfected with the indicated base editors and sgRNAs to insert (C and D) single or (E and F) double RMs or a scramble sgRNA (CTR). Cells were stimulated with 10 μ M forskolin before addition of iodide, indicated by the black arrow. (C and E) Representative traces of HS-YFP quenching assay recorded through fluorescence microscopy. (D and F) Quantification of CFTR function as $1 - F/F_0$ obtained 7 s from iodide addition. Data are means $\pm$ SDs from $n = 4$ independent experiments.

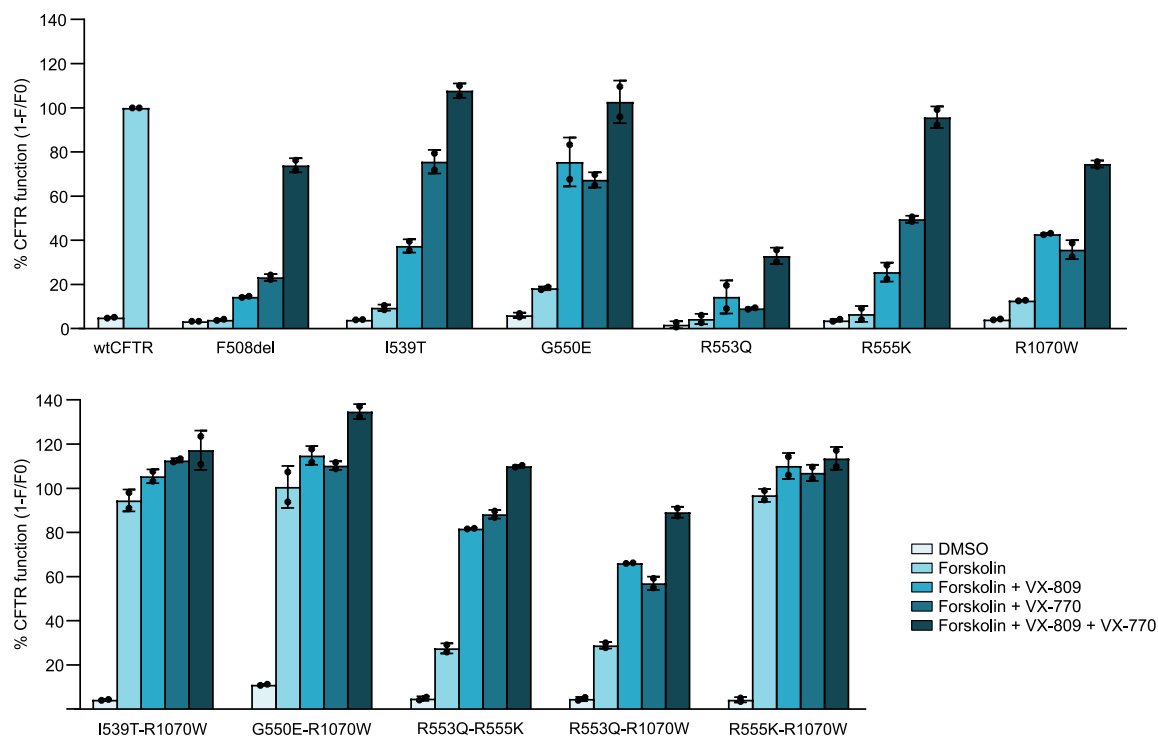


Figure 4. Evaluation of synergistic effects of RMs and modulators by HS-YFP quenching assay

Quantification of CFTR activity through HS-YFP quenching assay performed in HEK293T cells stably expressing F508del-CFTR variants and transfected with HS-YFP. Cells were treated either with forskolin, lumacaftor, or ivacaftor individually or in combination. CFTR function is expressed as $1 - F/F_0$ based on fluorescence values recorded 7 s after iodide addition. Data are means $\pm$ SDs from $n = 2$ independent experiments.

the inhibitor (DI-172) showed a 3- to 4-fold increase in CFTR function with respect to NT or CTR in both R1070W- and R555K-R1070W-edited cells. Conversely, no significant enhancement could be observed with the single R555K.

In epithelia modified with the R1070W and R1070W/R555K RMs, the CFTR current is approximately 1.13 and 1.78 μ A, which correspond to 3.39 and 5.34 μ A/cm² (Figure 6A). These data represent a 15%–20% and a 23%–31% activity rescue, respectively, with respect to an average CFTR function observed in non-CF epithelia under similar conditions (DI-172 = 17–23 μ A/cm²).^{53,54}

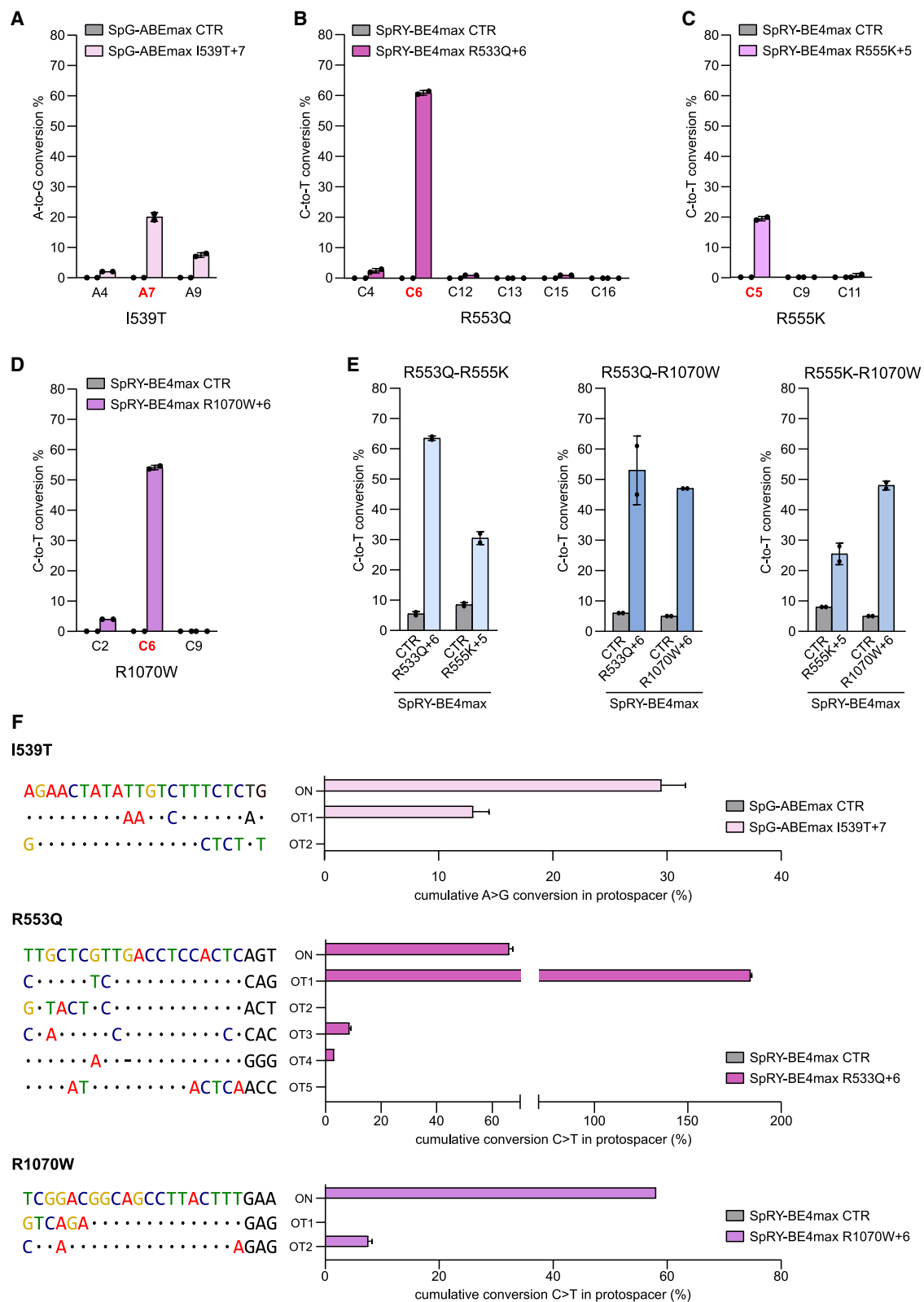
We also verified the potential additive effect that could be generated between RMs and ivacaftor, the potentiator molecule composing the ETI treatment. Ivacaftor, added after CPT-cAMP during Isc recordings, resulted in a further increase in CFTR-dependent current with a consequent enlargement of the effect of I-172 (Figure 6B). The effect of ivacaftor was particularly evident in cells edited with R1070W and R555K-R1070W. An increase in current by ivacaftor was also evident in traces from cells edited with R555K. We calculated DI-172 for all conditions. In this set of experiments, cells edited with R1070W and R555K-R1070W again showed an improvement in DI-172 value (3- to 4-fold increase compared to controls). There was also a positive trend for R555K cells, but this was not statistically significant (Figure 6B).

Finally, a third set of experiments was conducted with complete ETI by combining ivacaftor with the two correctors tezacaftor and elexacaftor. Under this condition, CFTR function was similarly rescued irrespective of the presence/absence of gene editing (Figure 6C). The lack of additive effects between pharmacological treatment and gene editing is likely due to the maximum F508del-CFTR rescue already obtained with ETI.

Overall, our data demonstrate that the insertion of RMs, specifically R1070W, in the F508del-CFTR locus of HBE cells efficiently rescues the CFTR ion-channel activity.

Finally, we also calculated the fraction of the total current that is activated by CPT-cAMP alone from the experiments performed in Figures 6B and 6C. The results, shown in Figure S9A, indicate that insertion of R1070W significantly increases this parameter compared to R555K. These results suggest that R1070W, besides improving the trafficking of mutant CFTR to the PM, also improves the gating as indicated by the proportionally reduced effect of ivacaftor.

We also evaluated the effect of gene editing on the effect of amiloride, added at the beginning of recordings. The amplitude of amiloride reflects not only the activity of the sodium channel ENaC but also in general the driving forces that underlie transepithelial ion transport.



(legend on next page)

As shown in Figure S9B, the amplitude of the current blocked by amiloride was not significantly affected by the electroporation procedure (CTR) or by gene editing.

DISCUSSION

Technological advancements in genome editing offer new opportunities for treating genetic diseases like CF. Several strategies have been developed to repair CF causing mutations including the F508del,⁹ the most frequent in pwCFs.¹ In several studies, the HDR capacity to substitute genomic fragments enhanced by nucleases was exploited to repair CF mutations, including the F508del.^{10–15} Nonetheless, despite relevant advancements, HDR remains a complex approach as it requires the delivery of a donor template and it is limited to dividing cells.⁵⁵ This poses significant obstacles for the treatment of the lung epithelium and potentially other tissues composed of differentiated and undifferentiated cells. Lungs are prioritized targets in CF as this is the organ mainly hampered by the disease. Natural barriers in human lungs further exacerbated in CF by the accumulation of thick and sticky mucus continue to pose a significant challenge for effective delivery to these organs.⁵⁶ Several viral and non-viral systems have been tested over the years, and recently, newly developed lipid nanoparticles, with an optimized formulation, have enabled targeted delivery of the CRISPR-Cas9 system to the lungs *in vivo*,^{57,58} prompting a significant advancement in the field of gene therapy for CF.

Nonetheless, the role of different cell types in the lung epithelium has not been defined yet, thus eluding the best cellular target for gene correction. Epithelial stem cells, while promising targets due to their fundamental role in tissue regeneration, present risks since nuclease activity associated with HDR and other genome editing strategies may negatively impact their staminal properties with potential downstream negative effects in derived tissues.^{59–61} Moreover, DSBs generated by nucleases in HDR and other approaches have been reported to induce genomic rearrangements, including translocation and chromosome aberrations.^{16–18} In recent years, more DSB free tools have been developed. Prime editing is an efficient genome editing strategy for both dividing and non-dividing cells and allows the introduction of all 12 possible types of point mutations, small insertions, and small deletions.¹⁹ This tool is composed of various elements requiring fine-tuning for efficient and precise editing. A first attempt to correct the F508del locus through prime editing showed an overall low efficiency and high levels of indels introduced on both alleles due to the additional nicking sgRNA.²⁰ More recently, an intensive optimization study produced an enhanced prime editing strategy to repair the

F508del mutation.^{20,21} The optimization comprises several modifications of the editor, including the co-expression of a non-physiological dominant-negative mismatch repair protein (MLH1dn), which resulted in the recovery of CFTR activity similar to levels obtained with pharmacological treatments. The correction of the mutation was associated with indels that, even though less than the previous prime editing or HDR approach, still are remarkably high (around 10% at the highest editing conditions).

Base editing has emerged as another promising strategy for correcting CF mutations.^{45,62–64} It is usually more efficient than prime editing^{20,21} and is better suited for delivery due to reduced complexity compared to HDR, requiring a co-delivery with donor DNA template, and prime editor, using two guide RNAs (prime editing gRNA and nicking gRNA) plus a cofactor (MLH1dn) to enhance the editing.

Nonetheless, base editors are limited to A>G or C>T transitions, making them unsuitable for correcting deletions like the F508del. To overcome this limitation, we explored the use of RMs to compensate for the F508del-CFTR defect.²⁴ We selected RMs compatible with a base editing strategy to be introduced into the F508del allele as single or combined mutations. I539T, G550E, R553Q, and R555K, located in the NBD1 domain, are mainly responsible for restoring the protein folding,^{26–31} while the R1070W at the TMD2 domain increases PM levels and also improves protein gating.^{24,30,32} The combination of R555K and R1070W mutations proved particularly effective, as they act at different stages of CFTR biogenesis and complement each other in restoring CFTR function. The success of this approach was further enhanced by an opportunistic bystander edit that introduced an additional beneficial mutation, R553Q,⁶⁵ resulting in a triple modification (R553Q-R555K-R1070W) using a single base editor and two sgRNAs.

A second bystander edit associated with this RM combination was observed with the sgRNA for R1070W producing a silent mutation thus very likely innocuous. Testing additional deaminase variants with a narrowed editing window^{66,67} and different Cas9 orthologs^{68–71} might enhance specificity, avoiding the modification of OT bases. However, we already observed a lowering of bystander edits to background levels in HBE cells, probably as a result of transient expression of base editors delivered as RNA, as reported before.⁴⁵ The OT profile for selected sgRNAs was quite heterogeneous with diverse percentages of non-specific modifications. Focusing on the R555K-R1070W RMs, we detected exclusively a non-specific edit in the intronic region of the *FLRT2* gene, which deserves further

Figure 5. Insertion of RMs in HBE cells using base editors and evaluation of editing precision

(A–D) Editing efficiency for the insertion of single RMs in HBE cells homozygous for the F508del-CFTR gene. Editing was quantified by deep sequencing analyses. HBE were electroporated with mRNA encoding the indicated base editors and chemically modified sgRNAs specific for the target nucleotide or a scramble sgRNA (CTR). The target nucleotides are highlighted in red; bystander nucleotides are in black. Data are means $\pm$ SDs from $n = 2$ independent experiments. (E) Histograms reporting the editing efficiency obtained for the insertion of RM combinations in HBE cells homozygous for the F508del-CFTR gene. Cells were electroporated with SpRY-CBE4max mRNA and two targeting sgRNAs or a scramble sgRNA (CTR). Editing was quantified by the EditR tool⁴⁷ on Sanger sequencing chromatograms. Data are means $\pm$ SDs from $n = 2$ independent experiments. (F) Deep sequencing analysis of off-targets (OTs) identified through GUIDE-seq analysis. Cumulative A>G or C>T base conversions inside the sgRNA spacer are reported. Data are means $\pm$ SDs from $n = 2$ independent experiments.

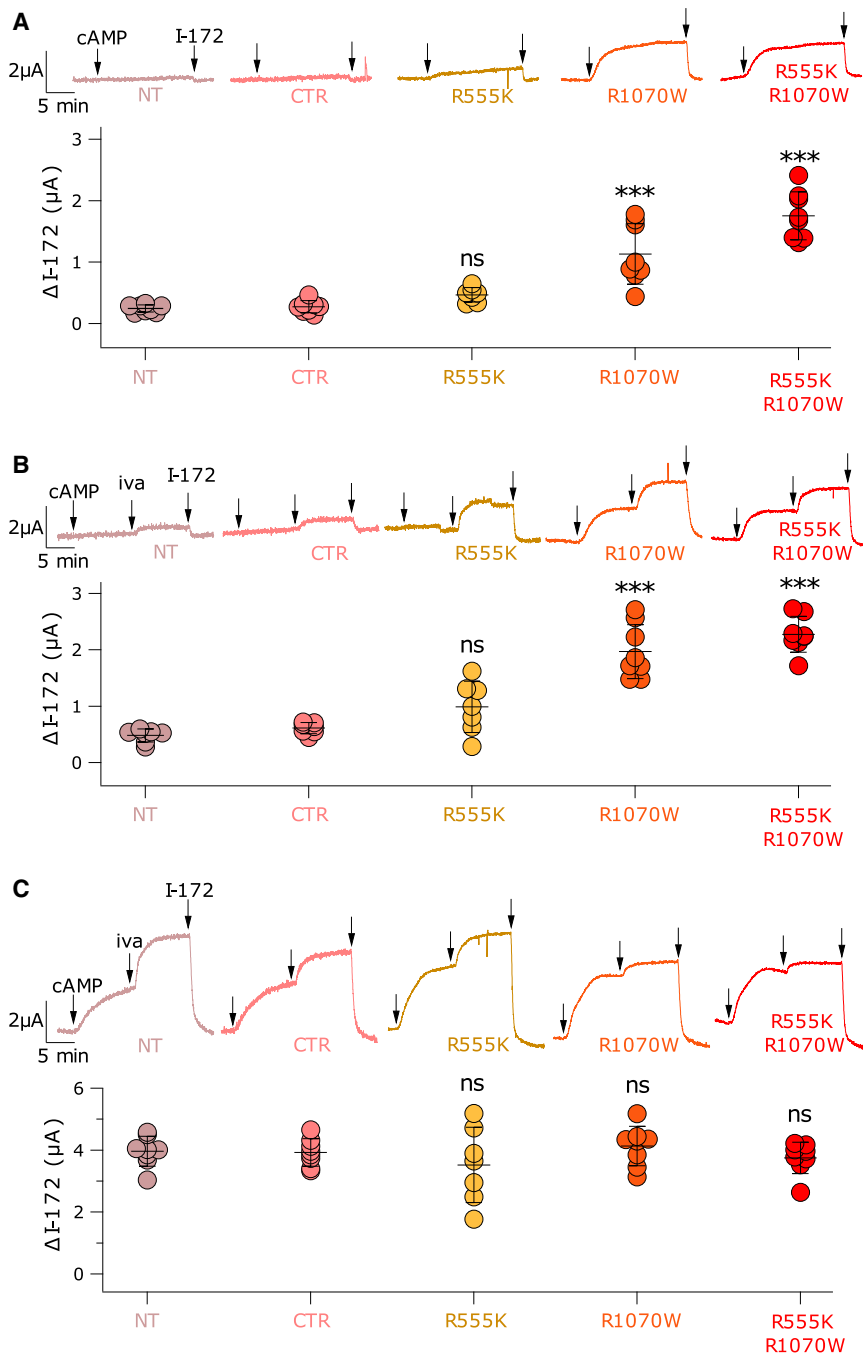


Figure 6. RMs insertion in HBE cells promote CFTR activity rescue

(A–C) Representative short-circuit current traces and summary of results obtained from epithelia generated from NT cells or cells treated with SpRY-CBE4max-CTR (scramble sgRNA) or SpRY-CBE4max-sgRNAs to insert the mutations. Editing efficiency is shown in Figure S8. For the representative traces, the identity of the compound addition (arrows) is shown for simplicity only in the far-left trace and the initial part of the recording, containing the effect of amiloride, has been removed. In each panel, the scatter dot plots report the amplitude of the current drop elicited by CFTRinh-172 (I-172). (A) Recordings with sequential addition of CPT-cAMP (cAMP) followed by I-172. (B) Recordings in which 1 μ M ivacaftor (iva) was added in the apical solution after CPT-cAMP. (C) Recordings in which epithelia were previously incubated with tezacaftor and ellexacaftor for 24 h. Scatter dot plots represent the results from epithelia of $n \geq 7$ of two independent experiments. Data are presented as means $\pm$ SDs.

genome editing strategy alone recovered CFTR activity, while it did not alter the sodium channel ENaC, as well as more general transepithelial ion transport, as observed by the response to the amiloride. A synergistic effect between the editing and pharmacological treatment was observed by using potentiators, molecules that improve the CFTR gating, in particular with R555K-R1070W or R1070W alone. These results indicate that the recovery of F508del-CFTR PM proximity by editing is sufficient to retrieve normal current flows by the potentiator treatment. Conversely, pharmacological cocktails containing both correctors and potentiators (ETI) reached the maximum enhancement of F508del-CFTR recovery, and no benefits could be generated by the RMs. Nonetheless, CFTR activity analyzed in cell lines expressing the F508del-CFTR modified with RMs showed an incremental effect of both correctors and potentiators with the majority of the editing strategies. These results suggest that increasing the editing efficacy through more advanced tools may lead to powerful combinatorial treatments

investigation to determine the impact in the target cells. The deep sequence analysis showed no indel formation at the target site, thus precluding genotoxic activity as observed for HDR and prime editing.

The capacity of the R555K-R1070W, introduced through base editing, to recover F508del-CFTR function was validated in terminally differentiated HBE cells derived from patients by measuring the short-circuit current in pseudostratified epithelia grown in culture. The

increasing the therapeutic efficacy of the CFTR modulators to lower the drug dosage regimen.

Even though CFTR modulators have significantly improved the quality of life of pwCFs and extended their lifespan, the burden of lifelong and costly treatments, which are also occasionally inefficacious, highlight the urgent need for a more definitive cure. Indeed, with this study, we first prove that beyond the value of a potential permanent

cure for CF, genome editing offers the opportunity to design a combinatorial treatment to increase the treatment solutions.

Harnessing the CFTR folding and activity by targeting specific domains is a new frontier for the treatment of CF.

METHODS

Plasmids

pCHMWS-CFTR-wt-ires-PURO and pCHMWS-CFTR-F508del-ires-PURO plasmids (kindly provided by Prof. Marianne Carlon, KU Leuven, Leuven, Belgium) contain a 3HA-tag in the fourth extracellular loop of CFTR and a puromycin resistance.

pLenti-CFTR-wt-ires-PURO and pLenti-CFTR-F508del-ires-PURO plasmids were cloned starting from a pLenti-FNLS-P2A-PURO plasmid. pLenti-FNLS-P2A-PURO was a gift from Lukas Dow (Addgene plasmid no. 110841; <http://n2t.net/addgene:110841>; RRID: Addgene_110841).⁵⁰ The BE4max-P2A-PURO sequence was substituted with CFTR-ires-PURO derived by PCR on the pCHMWS-CFTR plasmids, using BamHI and MluI restriction enzyme digestion.

F508del-CFTR RMs were cloned into the pLenti-CFTR-F508del-ires-PURO plasmid using PCR amplification and BsmBI restriction enzyme. The oligonucleotides used for cloning are listed in Table S1.

Plasmids expressing sgRNAs were cloned in a pUC19-sgRNAopt (pUC19 plasmid with sgRNA cassette derived from pX330) as described before.⁷² pX330-U6-Chimeric_BB-CBh-hSpCas9 was a gift from Feng Zhang (Addgene plasmid no. 42230; <http://n2t.net/addgene:42230>; RRID: Addgene_42230).⁷³ The sgRNA spacers are reported in Table S2.

The HS-YFP (YFP-H148Q/I152L/F46L) plasmid encoded for HS-YFP was used for the YFP assay.

In the base editing experiments, the following plasmids were cloned by restriction enzyme strategies: pCMV-SpG-ABEmax, pCMV-SpG-ABE8e, pCMV-SpG-ABE8.20m, pCMV-NG-ABEmax,⁴⁵ pCMV-NG-ABE8e,⁴⁵ pCMV-AncBE4max-SpG, pCAG-CBE4max-NG-P2A-EGFP, and pCMV-AncBE4max-NG. pCAG-CBE4max-SpRY-P2A-EGFP and pCAG-CBE4max-SpG-P2A-EGFP were a gift from Benjamin Kleinstiver (Addgene plasmid no. 139999; <http://n2t.net/addgene:139999>; RRID: Addgene_139999) (Addgene plasmid no. 139998; <http://n2t.net/addgene:139998>; RRID: Addgene_139998).⁴⁴

pIVT-CMV-SpG-ABEmaxP2A-EGFP and pIVT-CBE4max-SpRY-P2A-EGFP, used for *in vitro* RNA transcription, were constructed starting from pCMV-SpG-ABEmax and pCAG-CBE4max-SpRY-P2A-EGFP introducing the ABE/CBE-Cas9 sequence in the pIVT-SpRY-ABE9 plasmid by EcoRI restriction enzyme digestion. The oligonucleotides used for cloning are listed in Table S3.

For GUIDE-seq analyses, pCMV-T7-SpG-P2A-EGFP and pCMV-T7-SpRY-P2A-EGFP plasmids were used. pCMV-T7-SpG-P2A-EGFP and pCMV-T7-SpRY-P2A-EGFP were a gift from Benjamin Kleinstiver (Addgene plasmid no. 139988; <http://n2t.net/addgene:139988>; RRID: Addgene_139988) (Addgene plasmid no. 139989; <http://n2t.net/addgene:139989>; RRID: Addgene_139989).⁴⁴

Cell lines

HEK293T cells were obtained from the American Type Culture Collection (www.atcc.org). HEK293T stably expressing HS-YFP, wtCFTR, and F508del-CFTR were kindly provided by Prof. M.S. Carlon (KU Leuven, Leuven, Belgium). HEK293T cell lines stably expressing the different CFTR variants were generated by lentiviral transduction.

Cells were cultured in Dulbecco's modified Eagle's medium (Life Technologies) supplemented with 10% fetal bovine serum (Life Technologies), 10 U/mL antibiotics (penicillin-streptomycin; Life Technologies), and 2 mM L-glutamine at 37°C in a 5% CO₂ humidified atmosphere.

Lentiviral production and transduction of cell lines

Lentiviral particles were produced in HEK293T cells at 80% confluency in 100-mm dishes. A total of 10 µg pLenti-CFTR-F508del-RMs-ires-PURO plasmid, 6.5 µg Δ8.91 packaging plasmid, and 3.5 µg VSV-G were transfected using polyethylenimine. The viral supernatant was collected after 48 h and filtered in a 0.45-µm PES filter, and aliquots were stored at -80°C. Vector titers were measured as reverse transcriptase units (RTUs) by SYBR Green I-based qPCR-based product-enhanced reverse transcriptase method.⁷³

For transduction experiments, 150,000 HEK293T cells were seeded in a 24-well plate and transduced after 24 h with 1 RTU of lentiviral vectors. At 48 h later, cells were selected with puromycin (2 µg/mL). Cells were kept under puromycin selection, and experimental analyses were performed 3 weeks after transduction.

Cell line transfection

HEK293T cells or HEK293T stably expressing HS-YFP were co-transfected with 750 ng base editor and 250 ng sgRNA encoding plasmids (sgRNA scramble [CTR] and sgRNAs to insert I539T, R553Q, R555K, and R1070W). Transfections were performed using Transit LT1 reagent (Mirus Bio) according to the manufacturer's protocol. Cell pellets were collected 3 days post-transfection for editing analyses.

Editing analyses

Genomic DNA was extracted using QuickExtract DNA extraction solution (Epicentre). Target regions were amplified by PCR with HOT FIREPol DNA Polymerase (Solis BioDyne) using 100–500 ng DNA as template. Oligos are listed in Tables S4 and S5. PCR products were purified using NucleoSpin Gel and PCR Clean-up (Macherey-Nagel) and sequenced by Sanger sequencing. To evaluate editing efficiency, the chromatograms were analyzed using EditR software.⁴⁷

Western blotting

Cell pellets were collected from cells stably expressing CFTR variants and lysed with radioimmunoprecipitation assay buffer. A total of 20 µg of the whole lysates quantified by bicinchoninic acid assay (Thermo Fisher Scientific) were separated by SDS-PAGE in a 6% polyacrylamide gel and then transferred to a polyvinylidene fluoride membrane. Membranes were probed with a mix of four mouse monoclonal CFTR antibodies purchased from the CFF Foundation (Ab 570 and 596 diluted 1:500; Ab 660 and TJA9 diluted 1:1,000) and mouse anti- α -actinin (sc-17829, Santa Cruz, 1:1,000). Detection was performed with Pierce enhanced chemiluminescence reagent (Thermo Fisher Scientific) using the Uvitec Alliance Instrument.

Flow cytometry

Cells were stained with HA.11 antibody (no. 901515, BioLegend, 1:1,000) at 4°C for 45 min, followed by Alexa 488 secondary antibody (no. A-11001, Thermo Fisher Scientific, 1:500) for 30 min and then analyzed at BD FACS Canto, as previously described.⁷⁴ Cells not expressing CFTR were used as a negative control to determine the fraction of positive cells. wtCFTR and F508del-CFTR cells were used to determine the fraction of cells expressing CFTR at the PM (Figure S1).

HS-YFP assay

All transfections were performed using Transit LT1 reagent (Mirus Bio) according to the manufacturer's protocol. HEK293T stably expressing HS-YFP were transfected with 750 ng plasmids encoding CFTR wt, F508del, or F508del-RMs. After 24 h of transfection, cells were seeded into 96-well plates pre-coated with poly-D-lysine (no. A3890401, Thermo Fisher). After 24 h, HS-YFP quenching assays were conducted as follows.

Cells were washed with Dulbecco's PBS (DPBS; NaCl 137 mM, KCl 2.7 mM, Ca₂Cl 0.7 mM, MgCl₂ 1.1 mM, KH₂PO₄ 1.5 mM, Na₂HPO₄ 8.1 mM) and incubated in 60 µL DPBS containing either the CFTR activator forskolin (no. S2449 Selleckchem, 10 µM) or DMSO as a negative control for 20 min at 37°C. Fluorescence was measured using an inverted epifluorescence microscope equipped with a 490-nm excitation filter (Chroma Technology, ET490/20×) and a 540-nm emission filter. Continuous fluorescence recording was performed for 35 s. After an initial 5-s recording, 165 µL I⁻ buffer (NaI 137 mM, KCl 2.7 mM, CaCl 0.7 mM, MgCl₂ 1.1 mM, KH₂PO₄ 1.5 mM, Na₂PO₄ 8.1 mM) was added, and fluorescence was monitored for an additional 30 s. YFP quenching was calculated as the ratio F/F₀ at the end of the recording interval and CFTR function was expressed as 1 – (F/F₀).

For analysis of CFTR recovery induced by base editing, 150,000 HEK293T cells stably expressing HS-YFP were seeded into 24-well plates and transfected with 750 ng base editor and 250 ng sgRNA (sgRNA scramble [CTR] and sgRNAs to insert I539T, R553Q, R555K, and R1070W). At 24 h post-transfection, cells were moved into 96-well plates, and HS-YFP was performed after an additional 24 h.

To evaluate the synergistic effects between revertants and pharmacological modulators, HEK293T cells stably expressing CFTR variants were transfected with 250 ng plasmid encoding HS-YFP. The following day, cells were seeded in 96-well plates and incubated with either lumacaftor (no. S1565, Selleckchem, 2.5 µM) or DMSO for 24 h. After overnight incubation, cells were washed with DPBS, and a solution containing both the potentiator ivacaftor (no. S1144, Selleckchem, 3 µM) and forskolin (10 µM) were added for 20 min before starting the HS-YFP quenching assay.

In vitro mRNA transcription

SpG-ABEmax and SpRY-BE4max mRNAs were synthesized by T7 polymerase *in vitro* transcription using HighYield T7 RNA Synthesis Kit (Jena Bioscience). In detail, 750 ng linearized pIVT-SpG-ABEmax-EGFP and pIVT-SpRY-BE4max-EGFP plasmids were used as a template for the reaction. CleanCap Reagent M6 (TriLink BioTechnologies, 15 mM) was added to the reaction. mRNA was purified with LiCl precipitation and resuspended in diethylpyrocabonate-double-distilled H₂O. The sgRNAs (standard or with chemical modifications)⁷⁵ were synthetically ordered and produced from Integrated DNA Technologies (Table S6).

Electroporation of cell lines

A total of 200,000 HEK293T cells stably expressing F508del-CFTR were electroporated with 3 µg base editor mRNA either alone (control) or with 3 µg sgRNA. Electroporation was performed with the Amaxa 4D-Nucleofector (SE Cell Line buffer, CM-130 program). Next, cells were seeded into a 12-well plate and collected after 3 days for editing analysis.

HBE cell culture and electroporation

HBE cells were derived from a patient homozygous for the F508del and were provided by the Primary Cell Culture Service of the Italian Cystic Fibrosis Research Foundation. The Ethics Committee of the Istituto Giannina Gaslini approved this study, and informed consent was obtained from all participating CF subjects. Cells were cultured in LHC9/RPMI 1640 (1:1) without serum, as previously described,^{52,53} supplemented with rho-associated protein kinase 1 inhibitor (Y-27632, Merck, 5 mM), SMAD-signaling inhibitors, bone morphogenetic protein antagonist (DMH-1, Merck, 1 mM), and transforming growth factor β antagonist (A 83-01, Merck, 1 mM).⁷⁶ We electroporated 200,000 cells with the Amaxa 4D-Nucleofector (P3 primary cells buffer, DN-100 program) and seeded into a 12-well plate previously treated with collagen.

Several amounts of base editor mRNA and sgRNA R553Q+6 (standard or with chemical modifications)⁷⁵ were tested in electroporation experiments to find the most efficient condition (Figure S7). After the initial setting, the electroporation was performed for all the other revertants using 6 µg base editor mRNA either alone or with 6 µg sgRNA scramble (CTR) and sgRNAs to insert I539T, R553Q, R555K, and R1070W. After 3 days, cells were collected for editing analysis.

Short-circuit current recordings

HBE cells, resuspended in LHC9/RPMI 1640 proliferative medium, were seeded at high density on transwell (cc3470, Corning Costar, 0.33 cm² surface) porous inserts (150,000 cells per insert). After 24 h, to achieve the air-liquid interface condition, the medium on the apical side was totally removed, whereas the medium on the basolateral side was replaced with PneumaCult ALI (STEMCELL Technologies) differentiation medium. Short-circuit current experiments were performed after 2–3 weeks in culture, when epithelia were fully differentiated. Transwell supports carrying the epithelia were mounted in vertical chambers with internal fluid circulation (EM-CSYS-8, Physiologic Instruments). The apical and basolateral compartments were filled with a solution containing (in mM) 126 NaCl, 0.38 KH₂PO₄, 2.13 K₂HPO₄, 1 CaCl₂, 1 MgSO₄, 24 NaHCO₃, 10 glucose, and phenol red. Solutions were continuously bubbled with a mixture of 5% CO₂-95% air and kept at 37°C. The transepithelial voltage was clamped at 0 mV with an eight-channel voltage-clamp amplifier (VCC MC8, Physiologic Instruments) connected to apical and basolateral compartments with Ag/AgCl electrodes through agar bridges (1 M KCl in 2% agar). The resulting short-circuit current was recorded with the Acquire & Analyze 2.3 software (Physiologic Instruments). During experiments, epithelia were sequentially treated with amiloride (10 μM, apical) to block ENaC-dependent current, CPT-cAMP (100 μM, apical and basolateral) to stimulate CFTR activity, and CFTRinh-172 (20 μM, apical) for CFTR inhibition.

GUIDE-seq

GUIDE-seq analysis was performed as previously described.^{45,77} In short, HEK293 cells were transfected with 500 ng SpGCas9 or SpRYCas9 plasmids, 250 ng sgRNA plasmids, 10 pmol dsODNs and 50 ng pEGFP-IRES-PURO plasmid using Lipofectamine 3000 (Invitrogen). At 24 h after transfection, cells were split and selected with 1 mg/mL puromycin. Genomic DNA was extracted (DNeasy Blood and Tissue Kit, no. 69506, Qiagen) and sheared to an average length of 500 bp using sonication (Focused-ultrasonicator, Covaris). End-repair and sequencing adaptor ligation were performed using NEBNext Ultra End Repair/da Tailing Module (E7546S, New England Biolabs) and NEBNextUltra Ligation module (E7595S, New England Biolabs) following the procedures reported by Nobles et al.⁷⁸ Amplification steps were then performed following the GUIDE-seq protocol from Tsai et al.⁵¹ and libraries were quantified using the Qubit dsDNA High Sensitivity Assay kit (Q33233, Invitrogen). Next-generation sequencing was performed on an Illumina MiSeq sequencing device using Illumina Reagent kit V2-150PE. Raw sequencing data (FASTQ files) were analyzed using the GUIDE-seq computational pipeline (version 1.0.2) available at <https://github.com/tsailabSJ/guideseq>. The GUIDE-seq data are available in Table S7.

Targeted deep sequencing

Genomic DNA was extracted from HBE cells (F508del/F508del) 3 days after electroporation. The loci of interest (on-target and OT) were amplified using Phusion high-fidelity polymerase (Thermo Scientific). Amplicons were indexed by PCR using Nextera indexes

(Illumina), pooled in equimolar concentrations, and sequenced on an Illumina Miseq system using an Illumina Miseq Reagent kit V-300 cycles (2 × 150 bp paired end). Primers used for PCR amplification are listed in Table S8. FASTQ data were analyzed using CRISPResso version 2.3.1.⁷⁹

Statistical analysis

All statistical analyses were performed by GraphPad Prism 10 software. For HBE cells, ordinary one-way analysis of variance (ANOVA) was performed, using Dunnett's post hoc test. Statistical significance was defined as **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001, and ns, not significant.

DATA AND CODE AVAILABILITY

GUIDE-seq and targeted deep-sequencing data have been deposited at BioProject (<https://www.ncbi.nlm.nih.gov/bioproject/>) under the accession number PRJNA1150519. All other relevant data are available from the authors upon request.

ACKNOWLEDGMENTS

We are grateful to Dr. Cereseto's lab members for helpful discussions throughout the project. We thank the Next-Generation Sequencing facility at the University of Trento for technical support. We are grateful to Prof. Marianne Carlon for sharing reagents and for precious insights on CFTR testing. This work was fully supported by the Italian Cystic Fibrosis Foundation for Research (FFC) through FFC#3-2019, FFC#2-2021, and GenDel-CF for which we thank all the FFC delegations, FFC support groups and companies that have adopted them and by CARITRO Foundation grant no. 2018.256.

AUTHOR CONTRIBUTIONS

I.C., G.M., C.G., A.U., D.G., and M.D.S. designed and performed the experiments. I.C., G.M., C.G., A.U., D.G., M.D.S., L.J.V.G., and M.C. collected and analyzed the data. G.P., G.M., D.A., and A.C. conceived and designed the study. I.C., G.M., A.U., L.J.V.G., D.G., D.A., and A.C. wrote and edited the paper. A.C. and D.A. were responsible for the coordination of the study. All authors read, corrected, and approved the final manuscript.

DECLARATION OF INTERESTS

A.C. and G.P. are co-founders of and hold stock in Alia Therapeutics, a genome editing company.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.ymthe.2025.01.011>.

REFERENCES

- Ong, T., and Ramsey, B.W. (2023). Cystic Fibrosis: A Review. *JAMA* 329, 1859–1871. <https://doi.org/10.1001/jama.2023.8120>.
- Jia, S., and Taylor-Cousar, J.L. (2023). Cystic Fibrosis Modulator Therapies. *Annu. Rev. Med.* 74, 413–426. <https://doi.org/10.1146/annurev-med-042921-021447>.
- Cutting, G.R. (2015). Cystic fibrosis genetics: from molecular understanding to clinical application. *Nat. Rev. Genet.* 16, 45–56. <https://doi.org/10.1038/nrg3849>.
- The Clinical and Functional TRAnslation of CFTR (CFTR2) (US CF Foundation, Johns Hopkins University, The Hospital for Sick Children). <http://cfr2.org>.
- Ward, C.L., Omura, S., and Kopito, R.R. (1995). Degradation of CFTR by the ubiquitin-proteasome pathway. *Cell* 83, 121–127. [https://doi.org/10.1016/0092-8674\(95\)90240-6](https://doi.org/10.1016/0092-8674(95)90240-6).
- Sondo, E., Cresta, F., Pastorino, C., Tomati, V., Capurro, V., Pesce, E., Lena, M., Iacomino, M., Baffico, A.M., Coviello, D., et al. (2022). The L467F-F508del Complex Allele Hampers Pharmacological Rescue of Mutant CFTR by Elexacaftor/Tezacaftor/Ivacaftor in Cystic Fibrosis Patients: The Value of the Ex Vivo Nasal

- Epithelial Model to Address Non-Responders to CFTR-Modulating Drugs. *Int. J. Mol. Sci.* 23, 3175. <https://doi.org/10.3390/ijms23063175>.
7. Relling, M.V., and Evans, W.E. (2015). Pharmacogenomics in the clinic. *Nature* 526, 343–350. <https://doi.org/10.1038/nature15817>.
 8. Healey, N. (2024). Next-generation CRISPR-based gene-editing therapies tested in clinical trials. *Nat. Med.* 30, 2380–2381. <https://doi.org/10.1038/d41591-024-00056-8>.
 9. Maule, G., Arosio, D., and Cereseto, A. (2020). Gene Therapy for Cystic Fibrosis: Progress and Challenges of Genome Editing. *Int. J. Mol. Sci.* 21, 3903. <https://doi.org/10.3390/ijms21113903>.
 10. Lee, C.M., Flynn, R., Hollywood, J.A., Scallan, M.F., and Harrison, P.T. (2012). Correction of the $\Delta F508$ Mutation in the Cystic Fibrosis Transmembrane Conductance Regulator Gene by Zinc-Finger Nuclease Homology-Directed Repair. *Biores. Open Access* 1, 99–108. <https://doi.org/10.1089/biores.2012.0218>.
 11. Schwank, G., Koo, B.-K., Sasselli, V., Dekkers, J.F., Heo, I., Demircan, T., Sasaki, N., Boymans, S., Cuppen, E., van der Ent, C.K., et al. (2013). Functional repair of CFTR by CRISPR/Cas9 in intestinal stem cell organoids of cystic fibrosis patients. *Cell Stem Cell* 13, 653–658. <https://doi.org/10.1016/j.stem.2013.11.002>.
 12. Ruan, J., Hirai, H., Yang, D., Ma, L., Hou, X., Jiang, H., Wei, H., Rajagopalan, C., Mou, H., Wang, G., et al. (2019). Efficient Gene Editing at Major CFTR Mutation Loci. *Mol. Ther. Nucleic Acids* 16, 73–81. <https://doi.org/10.1016/j.omtn.2019.02.006>.
 13. Vaidyanathan, S., Salahudeen, A.A., Sellers, Z.M., Bravo, D.T., Choi, S.S., Batish, A., Le, W., Baik, R., de la O, S., Kaushik, M.P., et al. (2020). High-Efficiency, Selection-free Gene Repair in Airway Stem Cells from Cystic Fibrosis Patients Rescues CFTR Function in Differentiated Epithelia. *Cell Stem Cell* 26, 161–171.e4. <https://doi.org/10.1016/j.stem.2019.11.002>.
 14. Vaidyanathan, S., Baik, R., Chen, L., Bravo, D.T., Suarez, C.J., Abazari, S.M., Salahudeen, A.A., Dudek, A.M., Teran, C.A., Davis, T.H., et al. (2022). Targeted replacement of full-length CFTR in human airway stem cells by CRISPR-Cas9 for pan-mutation correction in the endogenous locus. *Mol. Ther.* 30, 223–237. <https://doi.org/10.1016/j.ymthe.2021.03.023>.
 15. Crane, A.M., Kramer, P., Bui, J.H., Chung, W.J., Li, X.S., Gonzalez-Garay, M.L., Hawkins, F., Liao, W., Mora, D., Choi, S., et al. (2015). Targeted correction and restored function of the CFTR gene in cystic fibrosis induced pluripotent stem cells. *Stem Cell Rep.* 4, 569–577. <https://doi.org/10.1016/j.stemcr.2015.02.005>.
 16. Leibowitz, M.L., Papatthanasou, S., Doerfler, P.A., Blaine, L.J., Sun, L., Yao, Y., Zhang, C.-Z., Weiss, M.J., and Pellman, D. (2021). Chromothripsis as an on-target consequence of CRISPR-Cas9 genome editing. *Nat. Genet.* 53, 895–905. <https://doi.org/10.1038/s41588-021-00838-7>.
 17. Nahmad, A.D., Reuveni, E., Goldschmidt, E., Tenne, T., Liberman, M., Horovitz-Fried, M., Khosravi, R., Kobo, H., Reinstein, E., Madi, A., et al. (2022). Frequent aneuploidy in primary human T cells after CRISPR–Cas9 cleavage. *Nat. Biotechnol.* 40, 1807–1813. <https://doi.org/10.1038/s41587-022-01377-0>.
 18. Kosicki, M., Tomberg, K., and Bradley, A. (2018). Repair of double-strand breaks induced by CRISPR-Cas9 leads to large deletions and complex rearrangements. *Nat. Biotechnol.* 36, 765–771. <https://doi.org/10.1038/nbt.4192>.
 19. Anzalone, A.V., Randolph, P.B., Davis, J.R., Sousa, A.A., Koblan, L.W., Levy, J.M., Chen, P.J., Wilson, C., Newby, G.A., Raguram, A., and Liu, D.R. (2019). Search-and-replace genome editing without double-strand breaks or donor DNA. *Nature* 576, 149–157. <https://doi.org/10.1038/s41586-019-1711-4>.
 20. Geurts, M.H., de Poel, E., Pleguezuelos-Manzano, C., Oka, R., Carrillo, L., Andersson-Rolf, A., Boretto, M., Brunsveld, J.E., van Boxtel, R., Beekman, J.M., and Clevers, H. (2021). Evaluating CRISPR-based prime editing for cancer modeling and CFTR repair in organoids. *Life Sci. Alliance* 4, e202000940. <https://doi.org/10.26508/lsa.202000940>.
 21. Sousa, A.A., Hemez, C., Lei, L., Traore, S., Kulhankova, K., Newby, G.A., Doman, J.L., Oye, K., Pandey, S., Karp, P.H., et al. (2025). Systematic optimization of prime editing for the efficient functional correction of CFTR F508del in human airway epithelial cells. *Nat. Biomed. Eng.* 9, 7–21. <https://doi.org/10.1038/s41551-024-01233-3>.
 22. Lino, C.A., Harper, J.C., Carney, J.P., and Timlin, J.A. (2018). Delivering CRISPR: a review of the challenges and approaches. *Drug Deliv.* 25, 1234–1257. <https://doi.org/10.1080/10717544.2018.1474964>.
 23. van Haasteren, J., Li, J., Scheideler, O.J., Murthy, N., and Schaffer, D.V. (2020). The delivery challenge: fulfilling the promise of therapeutic genome editing. *Nat. Biotechnol.* 38, 845–855. <https://doi.org/10.1038/s41587-020-0565-5>.
 24. Prins, S., Corradi, V., Sheppard, D.N., Tieleman, D.P., and Vergani, P. (2022). Can two wrongs make a right? F508del-CFTR ion channel rescue by second-site mutations in its transmembrane domains. *J. Biol. Chem.* 298, 101615. <https://doi.org/10.1016/j.jbc.2022.101615>.
 25. Dörk, T., Wulbrand, U., Richter, T., Neumann, T., Wolfes, H., Wulf, B., Maass, G., and Tümmler, B. (1991). Cystic fibrosis with three mutations in the cystic fibrosis transmembrane conductance regulator gene. *Hum. Genet.* 87, 441–446. <https://doi.org/10.1007/BF00197165>.
 26. DeCarvalho, A.C.V., Gansheroff, L.J., and Teem, J.L. (2002). Mutations in the nucleotide binding domain 1 signature motif region rescue processing and functional defects of cystic fibrosis transmembrane conductance regulator delta f508. *J. Biol. Chem.* 277, 35896–35905. <https://doi.org/10.1074/jbc.M205644200>.
 27. Teem, J.L., Carson, M.R., and Welsh, M.J. (1996). Mutation of R555 in CFTR-delta F508 enhances function and partially corrects defective processing. *Recept. Channels* 4, 63–72.
 28. He, L., Aleksandrov, L.A., Cui, L., Jensen, T.J., Nesbitt, K.L., and Riordan, J.R. (2010). Restoration of domain folding and interdomain assembly by second-site suppressors of the DeltaF508 mutation in CFTR. *FASEB J.* 24, 3103–3112. <https://doi.org/10.1096/fj.09-141788>.
 29. Rabeh, W.M., Bossard, F., Xu, H., Okiyoneda, T., Bagdany, M., Mulvihill, C.M., Du, K., di Bernardo, S., Liu, Y., Konermann, L., et al. (2012). Correction of both NBD1 energetics and domain interface is required to restore $\Delta F508$ CFTR folding and function. *Cell* 148, 150–163. <https://doi.org/10.1016/j.cell.2011.11.024>.
 30. Thibodeau, P.H., Richardson, J.M., Wang, W., Millen, L., Watson, J., Mendoza, J.L., Du, K., Fischman, S., Senderowitz, H., Lukacs, G.L., et al. (2010). The Cystic Fibrosis-causing Mutation $\Delta F508$ Affects Multiple Steps in Cystic Fibrosis Transmembrane Conductance Regulator Biogenesis. *J. Biol. Chem.* 285, 35825–35835. <https://doi.org/10.1074/jbc.M110.131623>.
 31. Mendoza, J.L., Schmidt, A., Li, Q., Nuvaga, E., Barrett, T., Bridges, R.J., Feranchak, A.P., Brautigam, C.A., and Thomas, P.J. (2012). Requirements for efficient correction of $\Delta F508$ CFTR revealed by analyses of evolved sequences. *Cell* 148, 164–174. <https://doi.org/10.1016/j.cell.2011.11.023>.
 32. Okiyoneda, T., Veit, G., Dekkers, J.F., Bagdany, M., Soya, N., Xu, H., Roldan, A., Verkman, A.S., Kurth, M., Simon, A., et al. (2013). Mechanism-based corrector combination restores $\Delta F508$ -CFTR folding and function. *Nat. Chem. Biol.* 9, 444–454. <https://doi.org/10.1038/nchembio.1253>.
 33. O'Riordan, C.R., Lachapelle, A.L., Marshall, J., Higgins, E.A., and Cheng, S.H. (2000). Characterization of the oligosaccharide structures associated with the cystic fibrosis transmembrane conductance regulator. *Glycobiology* 10, 1225–1233. <https://doi.org/10.1093/glycob/10.11.1225>.
 34. Farinha, C.M., Santos, L., and Ferreira, J.F. (2024). Cell type-specific regulation of CFTR trafficking-on the verge of progress. *Front. Cell Dev. Biol.* 12, 1338892. <https://doi.org/10.3389/fcell.2024.1338892>.
 35. Vijftigschild, L.A.W., van der Ent, C.K., and Beekman, J.M. (2013). A novel fluorescent sensor for measurement of CFTR function by flow cytometry. *Cytometry. A* 83, 576–584. <https://doi.org/10.1002/cyto.a.22275>.
 36. Galletta, L.V., Jayaraman, S., and Verkman, A.S. (2001). Cell-based assay for high-throughput quantitative screening of CFTR chloride transport agonists. *Am. J. Physiol. Cell Physiol.* 281, C1734–C1742. <https://doi.org/10.1152/ajpcell.2001.281.5.C1734>.
 37. Rees, H.A., and Liu, D.R. (2018). Base editing: precision chemistry on the genome and transcriptome of living cells. *Nat. Rev. Genet.* 19, 770–788. <https://doi.org/10.1038/s41576-018-0059-1>.
 38. Gaudelli, N.M., Komor, A.C., Rees, H.A., Packer, M.S., Badran, A.H., Bryson, D.I., and Liu, D.R. (2017). Programmable base editing of A·T to G·C in genomic DNA without DNA cleavage. *Nature* 551, 464–471. <https://doi.org/10.1038/nature24644>.
 39. Komor, A.C., Kim, Y.B., Packer, M.S., Zuris, J.A., and Liu, D.R. (2016). Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage. *Nature* 533, 420–424. <https://doi.org/10.1038/nature17946>.

40. Gaudelli, N.M., Lam, D.K., Rees, H.A., Solá-Esteves, N.M., Barrera, L.A., Born, D.A., Edwards, A., Gehrke, J.M., Lee, S.-J., Liguori, A.J., et al. (2020). Directed evolution of adenine base editors with increased activity and therapeutic application. *Nat. Biotechnol.* 38, 892–900. <https://doi.org/10.1038/s41587-020-0491-6>.
41. Koblan, L.W., Doman, J.L., Wilson, C., Levy, J.M., Tay, T., Newby, G.A., Maianti, J.P., Raguram, A., and Liu, D.R. (2018). Improving cytidine and adenine base editors by expression optimization and ancestral reconstruction. *Nat. Biotechnol.* 36, 843–846. <https://doi.org/10.1038/nbt.4172>.
42. Richter, M.F., Zhao, K.T., Eton, E., Lapinaite, A., Newby, G.A., Thuronyi, B.W., Wilson, C., Koblan, L.W., Zeng, J., Bauer, D.E., et al. (2020). Phage-assisted evolution of an adenine base editor with improved Cas domain compatibility and activity. *Nat. Biotechnol.* 38, 883–891. <https://doi.org/10.1038/s41587-020-0453-z>.
43. Nishimasu, H., Shi, X., Ishiguro, S., Gao, L., Hirano, S., Okazaki, S., Noda, T., Abudayyeh, O.O., Gootenberg, J.S., Mori, H., et al. (2018). Engineered CRISPR-Cas9 nuclease with expanded targeting space. *Science* 361, 1259–1262. <https://doi.org/10.1126/science.aas9129>.
44. Walton, R.T., Christie, K.A., Whittaker, M.N., and Kleinstiver, B.P. (2020). Unconstrained genome targeting with near-PAMless engineered CRISPR-Cas9 variants. *Science* 368, 290–296. <https://doi.org/10.1126/science.aba8853>.
45. Amistadi, S., Maule, G., Ciciani, M., Ensink, M.M., De Keersmaecker, L., Ramalho, A.S., Guidone, D., Buccicrossi, M., Galletta, L.J.V., Carlon, M.S., and Cereseto, A. (2023). Functional restoration of a CFTR splicing mutation through RNA delivery of CRISPR adenine base editor. *Mol. Ther.* 31, 1647–1660. <https://doi.org/10.1016/j.ymthe.2023.03.004>.
46. Veit, G., Da Fonte, D.F., Avramescu, R.G., Premchandrar, A., Bagdany, M., Xu, H., Bensinger, D., Stubba, D., Schmidt, B., Matouk, E., and Lukacs, G.L. (2020). Mutation-specific dual potentiators maximize rescue of CFTR gating mutants. *J. Cyst. Fibros.* 19, 236–244. <https://doi.org/10.1016/j.jcf.2019.10.011>.
47. Kluesner, M.G., Nedveck, D.A., Lahr, W.S., Garbe, J.R., Abrahante, J.E., Webber, B.R., and Moriarity, B.S. (2018). EditR: A Method to Quantify Base Editing from Sanger Sequencing. *CRISPR J.* 1, 239–250. <https://doi.org/10.1089/crispr.2018.0014>.
48. Antoniou, P., Hardouin, G., Martinucci, P., Frati, G., Felix, T., Chalumeau, A., Fontana, L., Martin, J., Masson, C., Brusson, M., et al. (2022). Base-editing-mediated dissection of a γ -globin cis-regulatory element for the therapeutic reactivation of fetal hemoglobin expression. *Nat. Commun.* 13, 6618–6622. <https://doi.org/10.1038/s41467-022-34493-1>.
49. Gehrke, J.M., Cervantes, O., Clement, M.K., Wu, Y., Zeng, J., Bauer, D.E., Pinello, L., and Joung, J.K. (2018). An APOBEC3A-Cas9 base editor with minimized bystander and off-target activities. *Nat. Biotechnol.* 36, 977–982. <https://doi.org/10.1038/nbt.4199>.
50. Zafra, M.P., Schatoff, E.M., Katti, A., Foronda, M., Breinig, M., Schweitzer, A.Y., Simon, A., Han, T., Goswami, S., Montgomery, E., et al. (2018). Optimized base editors enable efficient editing in cells, organoids and mice. *Nat. Biotechnol.* 36, 888–893. <https://doi.org/10.1038/nbt.4194>.
51. Tsai, S.Q., Zheng, Z., Nguyen, N.T., Liebers, M., Topkar, V.V., Thapar, V., Wyvekens, N., Khayter, C., Iafrate, A.J., Le, L.P., et al. (2015). GUIDE-seq enables genome-wide profiling of off-target cleavage by CRISPR-Cas nucleases. *Nat. Biotechnol.* 33, 187–197. <https://doi.org/10.1038/nbt.3117>.
52. Scudieri, P., Caci, E., Bruno, S., Ferrera, L., Schiavon, M., Sondo, E., Tomati, V., Gianotti, A., Zegarra-Moran, O., Pedemonte, N., et al. (2012). Association of TMEM16A chloride channel overexpression with airway goblet cell metaplasia. *J. Physiol.* 590, 6141–6155. <https://doi.org/10.1113/jphysiol.2012.240838>.
53. Scudieri, P., Musante, I., Venturini, A., Guidone, D., Genovese, M., Cresta, F., Caci, E., Palleschi, A., Poeta, M., Santamaria, F., et al. (2020). Ionocytes and CFTR Chloride Channel Expression in Normal and Cystic Fibrosis Nasal and Bronchial Epithelial Cells. *Cells* 9, 2090. <https://doi.org/10.3390/cells9092090>.
54. Capurro, V., Tomati, V., Sondo, E., Renda, M., Borrelli, A., Pastorino, C., Guidone, D., Venturini, A., Giraudo, A., Mandrup Bertozzi, S., et al. (2021). Partial rescue of F508del-CFTR stability and trafficking defects by double corrector treatment. *Int. J. Mol. Sci.* 22, 5262. <https://doi.org/10.3390/ijms22105262>.
55. Mao, Z., Bozzella, M., Seluanov, A., and Gorbunova, V. (2008). DNA repair by nonhomologous end joining and homologous recombination during cell cycle in human cells. *Cell Cycle* 7, 2902–2906. <https://doi.org/10.4161/cc.7.18.6679>.
56. Zierden, H.C., Josyula, A., Shapiro, R.L., Hsueh, H.T., Hanes, J., and Ensign, L.M. (2021). Avoiding a sticky situation: Bypassing the mucus barrier for improved local drug delivery. *Trends Mol. Med.* 27, 436–450. <https://doi.org/10.1016/j.molmed.2020.12.001>.
57. Chen, K., Han, H., Zhao, S., Xu, B., Yin, B., Trinidad, M., Burgstone, B.W., Murthy, N., and Doudna, J.A. (2023). Lung and liver editing by lipid nanoparticle delivery of a stable CRISPR-Cas9 RNP. Preprint at bioRxiv. <https://doi.org/10.1101/2023.11.15.566339>.
58. Wang, X., Liu, S., Sun, Y., Yu, X., Lee, S.M., Cheng, Q., Wei, T., Gong, J., Robinson, J., Zhang, D., et al. (2023). Preparation of selective organ-targeting (SORT) lipid nanoparticles (LNPs) using multiple technical methods for tissue-specific mRNA delivery. *Nat. Protoc.* 18, 265–291. <https://doi.org/10.1038/s41596-022-00755-x>.
59. Zhang, Z., Bao, X., and Lin, C.-P. (2023). Progress and Prospects of Gene Editing in Pluripotent Stem Cells. *Biomedicine* 11, 2168. <https://doi.org/10.3390/biomedicine11082168>.
60. Ihry, R.J., Worringer, K.A., Salick, M.R., Frias, E., Ho, D., Theriault, K., Kommineni, S., Chen, J., Sondey, M., Ye, C., et al. (2018). p53 inhibits CRISPR-Cas9 engineering in human pluripotent stem cells. *Nat. Med.* 24, 939–946. <https://doi.org/10.1038/s41591-018-0050-6>.
61. Schirotti, G., Conti, A., Ferrari, S., Della Volpe, L., Jacob, A., Albano, L., Beretta, S., Calabria, A., Vavassori, V., Gasparini, P., et al. (2019). Precise Gene Editing Preserves Hematopoietic Stem Cell Function following Transient p53-Mediated DNA Damage Response. *Cell Stem Cell* 24, 551–565.e8. <https://doi.org/10.1016/j.stem.2019.02.019>.
62. Mention, K., Cavusoglu-Doran, K., Joynt, A.T., Santos, L., Sanz, D., Eastman, A.C., Merlo, C., Langfelder-Schwind, E., Scallan, M.F., Farinha, C.M., et al. (2023). Use of adenine base editing and homology-independent targeted integration strategies to correct the cystic fibrosis causing variant. *Hum. Mol. Genet.* 32, 3237–3248. <https://doi.org/10.1093/hmg/ddad143>.
63. Geurts, M.H., de Poel, E., Amatngalim, G.D., Oka, R., Meijers, F.M., Kruisselbrink, E., van Mourik, P., Berkers, G., de Winter-de Groot, K.M., Michel, S., et al. (2020). CRISPR-Based Adenine Editors Correct Nonsense Mutations in a Cystic Fibrosis Organoid Biobank. *Cell Stem Cell* 26, 503–510.e7. <https://doi.org/10.1016/j.stem.2020.01.019>.
64. Krishnamurthy, S., Traore, S., Cooney, A.L., Brommel, C.M., Kulhankova, K., Sinn, P.L., Newby, G.A., Liu, D.R., and McCray, P.B. (2021). Functional correction of CFTR mutations in human airway epithelial cells using adenine base editors. *Nucleic Acids Res.* 49, 10558–10572. <https://doi.org/10.1093/nar/gkab788>.
65. Teem, J.L., Berger, H.A., Ostedgaard, L.S., Rich, D.P., Tsui, L.C., and Welsh, M.J. (1993). Identification of revertants for the cystic fibrosis delta F508 mutation using STE6-CFTR chimeras in yeast. *Cell* 73, 335–346. [https://doi.org/10.1016/0092-8674\(93\)90233-g](https://doi.org/10.1016/0092-8674(93)90233-g).
66. Kim, D., Lim, K., Kim, S.-T., Yoon, S.-H., Kim, K., Ryu, S.-M., and Kim, J.-S. (2017). Genome-wide target specificities of CRISPR RNA-guided programmable deaminases. *Nat. Biotechnol.* 35, 475–480. <https://doi.org/10.1038/nbt.3852>.
67. Anzalone, A.V., Koblan, L.W., and Liu, D.R. (2020). Genome editing with CRISPR-Cas nucleases, base editors, transposases and prime editors. *Nat. Biotechnol.* 38, 824–844. <https://doi.org/10.1038/s41587-020-0561-9>.
68. Pedrazzoli, E., Demozzi, M., Visentin, E., Ciciani, M., Bonuzzi, I., Pezzè, L., Lucchetta, L., Maule, G., Amistadi, S., Esposito, F., et al. (2024). CoCas9 is a compact nuclease from the human microbiome for efficient and precise genome editing. *Nat. Commun.* 15, 3478. <https://doi.org/10.1038/s41467-024-47800-9>.
69. Fujii, W., Kakuta, S., Yoshioka, S., Kyuwa, S., Sugiura, K., and Naito, K. (2016). Zygote-mediated generation of genome-modified mice using *Streptococcus thermophilus* 1-derived CRISPR/Cas system. *Biochem. Biophys. Res. Commun.* 477, 473–476. <https://doi.org/10.1016/j.bbrc.2016.06.070>.
70. Ran, F.A., Cong, L., Yan, W.X., Scott, D.A., Gootenberg, J.S., Kriz, A.J., Zetsche, B., Shalem, O., Wu, X., Makarova, K.S., et al. (2015). In vivo genome editing using *Staphylococcus aureus* Cas9. *Nature* 520, 186–191. <https://doi.org/10.1038/nature14299>.
71. Edraki, A., Mir, A., Ibrahim, R., Gainetdinov, I., Yoon, Y., Song, C.-Q., Cao, Y., Gallant, J., Xue, W., Rivera-Pérez, J.A., and Sontheimer, E.J. (2019). A compact, high-accuracy Cas9 with a dinucleotide PAM for in vivo genome editing. *Mol. Cell* 73, 714–726.e4. <https://doi.org/10.1016/j.molcel.2018.12.003>.

72. Cong, L., Ran, F.A., Cox, D., Lin, S., Barretto, R., Habib, N., Hsu, P.D., Wu, X., Jiang, W., Marraffini, L.A., and Zhang, F. (2013). Multiplex genome engineering using CRISPR/Cas systems. *Science* 339, 819–823. <https://doi.org/10.1126/science.1231143>.
73. Petris, G., Casini, A., Montagna, C., Lorenzin, F., Prandi, D., Romanel, A., Zasso, J., Conti, L., Demichelis, F., and Cereseto, A. (2017). Hit and go CAS9 delivered through a lentiviral based self-limiting circuit. *Nat. Commun.* 8, 15334. <https://doi.org/10.1038/ncomms15334>.
74. Ensink, M., De Keersmaecker, L., Heylen, L., Ramalho, A.S., Gijssbers, R., Farré, R., De Boeck, K., Christ, F., Debyser, Z., and Carlon, M.S. (2020). Phenotyping of Rare CFTR Mutations Reveals Distinct Trafficking and Functional Defects. *Cells* 9, 754. <https://doi.org/10.3390/cells9030754>.
75. Riesenberger, S., Helmbrecht, N., Kanis, P., Maricic, T., and Pääbo, S. (2022). Improved gRNA secondary structures allow editing of target sites resistant to CRISPR-Cas9 cleavage. *Nat. Commun.* 13, 489. <https://doi.org/10.1038/s41467-022-28137-7>.
76. Mou, H., Vinarsky, V., Tata, P.R., Brazauskas, K., Choi, S.H., Crooke, A.K., Zhang, B., Solomon, G.M., Turner, B., Bihler, H., et al. (2016). Dual SMAD Signaling Inhibition Enables Long-Term Expansion of Diverse Epithelial Basal Cells. *Cell Stem Cell* 19, 217–231. <https://doi.org/10.1016/j.stem.2016.05.012>.
77. Maule, G., Casini, A., Montagna, C., Ramalho, A.S., De Boeck, K., Debyser, Z., Carlon, M.S., Petris, G., and Cereseto, A. (2019). Allele specific repair of splicing mutations in cystic fibrosis through AsCas12a genome editing. *Nat. Commun.* 10, 3556. <https://doi.org/10.1038/s41467-019-11454-9>.
78. Nobles, C.L., Reddy, S., Salas-McKee, J., Liu, X., June, C.H., Melenhorst, J.J., Davis, M.M., Zhao, Y., and Bushman, F.D. (2019). iGUIDE: an improved pipeline for analyzing CRISPR cleavage specificity. *Genome Biol.* 20, 14. <https://doi.org/10.1186/s13059-019-1625-3>.
79. Clement, K., Rees, H., Canver, M.C., Gehrke, J.M., Farouni, R., Hsu, J.Y., Cole, M.A., Liu, D.R., Joung, J.K., Bauer, D.E., and Pinello, L. (2019). CRISPResso2 provides accurate and rapid genome editing sequence analysis. *Nat. Biotechnol.* 37, 224–226. <https://doi.org/10.1038/s41587-019-0032-3>.